

# Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Digital System Design and Computer Organization [ECM\_2122]

Marks: 50

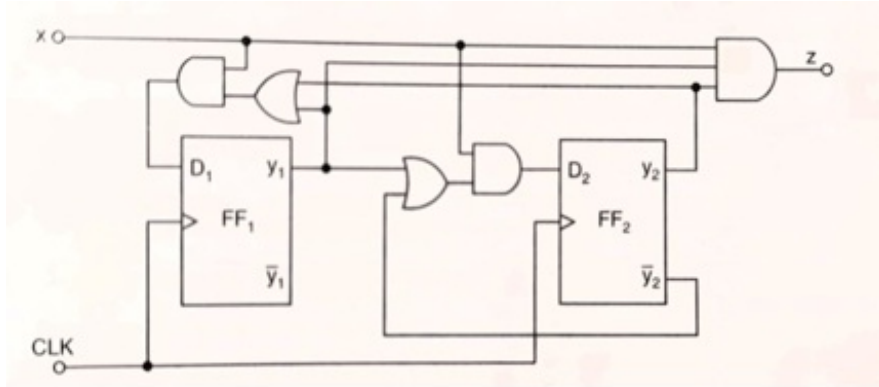
Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Design a combinational circuit using programmable logic array (PLA) to implement a full subtractor. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Construct a Mod-10 ripple up-counter using negative edge triggered T flip-flops and draw the output timing diagrams. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Write Verilog dataflow code to implement 8:1 multiplexer without using the instances of 2:1 multiplexers. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 2) Design a 3-bit self-correcting synchronous counter using JK flip-flops that follows the sequence 7-6-4-3 repeatedly, ensuring that any unused states automatically go to the state-7 on the next clock pulse. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Explain the Time Division Multiple Access bus arbitration protocol. (3)
- Course Outcome: "CO5" / Cognitive Level: "Understand"
- C) With an example, explain the role of instances in Verilog programming. (2)
- Course Outcome: "CO3" / Cognitive Level: "Understand"
- 3) Design a Mealy synchronous sequential circuit using D flip-flops with a serial input x and output z that produces  $z = 1$  whenever the input completes the sequence 1011. Overlapping is allowed. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Use the Booth's algorithm to multiply +23 (multiplicand) by -10 (multiplier), where each number is represented using 6 bits. (3)
- Course Outcome: "CO5" / Cognitive Level: "Application"
- C) Write a behavioral Verilog code for a 4:1 MUX using an "always block" and "case statement". (2)

- 4) Analyze the sequential circuit shown below and obtain its state table and state diagram. (4)



- B) Express the given statement in three, two and one instruction addressing formats:  $D = A + B \times C$  (4)

- C) What is the difference between non-pipelined and pipelined processors? If the number of stages is 6 and instructions to be executed are 5, how many cycles for execution are required in both cases? (2)

- 5) A mod-6 asynchronous up-counter using negative edge triggered T flip-flops, which resets itself if any undesired state occurs needs to be designed. Write the Verilog code to implement the same. (4)

- B) Define the following addressing modes with an example for each type: (4)
- (i) register-indirect, (ii) displacement, (iii) autodecrement, (iv) implied (implicit)

- C) Explain the operation of accumulator-based computer organization. (2)

-----End-----

| Event Name                        | Subject                                                  |
|-----------------------------------|----------------------------------------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025-2026 | ECM_2122:Digital System Design and Computer Organization |

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 4         |
| 2            | 2              | CO 2        | 17        |
| 3            | 3              | CO 3        | 11        |
| 4            | 4              | CO 4        | 10        |
| 5            | 5              | CO 5        | 8         |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name                    | Subject                                     |
|-------------------------------|---------------------------------------------|
| 058-MIT BLR-MAKEUP EXAM-2025- | ECM 2122:Digital System Design and Computer |

| # | Level       | No of Questions | Marks (%) |
|---|-------------|-----------------|-----------|
| 1 | Remember    | 0               | 0(0%)     |
| 2 | Understand  | 5               | 13(26%)   |
| 3 | Application | 10              | 37(74%)   |
| 4 | Analysis    | 0               | 0(0%)     |
| 5 | Create      | 0               | 0(0%)     |
| 6 | Evaluate    | 0               | 0(0%)     |
|   | Total       | 15              | 50 (100%) |